HR ATTRITION ANALYTICS DASHBOARD USING POWER BI

Project overview

Our team "Group No.4" consists of 6 colleges: -

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And we collaborate to complete this project properly to reach the highest standard.

So, **firstly**, we contact the responsible person in the database and the archive department in our company to **get data** as a premium phase and to make sure that we fulfill all the gaps for avoiding any missing data we already have, and make it as precise as we can before we establish and go forward in our analysis as the first step.

Secondly, we determined that our topic needs to apply the methodology of an operational Power BI dashboard and, for that purpose, apply all applicable KPIs measure which are directly related to the core issue for our HR department, which is confined to the turnover rate and how to figure out the main cause of this problem.

Thirdly, we must figure out the relations between the attrition rate and the other factors in our analysis.

Fourthly, we make sure to analyze the other factors that affect the main problem, such as the satisfaction rate and all employees' capabilities & data, which can be useful to our analysis.

Fifthly, we apply onto our analysis all formal and theoretical steps to our data in the Power Query to make sure it's precise and accurate before using it in our analysis to generate an interactive Power BI dashboard, beginning with: -

The first phase is the **cleaning phase**, which consists of handling all errors / replacing values, removing other blank columns, removing blank rows, removing duplicates, using the first row as a header, removing the top rows if it's not contained the headers, or applying the first row as a header.

the second phase as **data transformations** as ensure the columns header data type is correct or change it ensure the date columns are applying the local time without any errors or mistake or different style, renaming the columns header to a clear one and merged the employees name columns " the first name and the second name columns into one columns, applying the fill down approaches to all unmerged date especially in to the city data columns, etc.

The third phase is applying **data modeling** and making relations between the tables, so that in this phase, we already have one table containing all the information we need, which could be applied to our analysis.

The fourth phase is about **data visualizations** and creating the charts and graphs to convert all data we analyze into easy interactive charts could be used in data showing or presentations in front of our CEOs and in this phase we made a 4 slide dashboard represent all analysis about the main issue " one for attrition and one for employees satisfactions, and one for all date related to the employees such as performance, educational level, age, department, management rank, experience, etc.. and the last slide are about all company overview and the red alarms indicators which is alerting us about the what may be happened in the nearest future} and in this umpteenth phase we take into our consideration that our the 4 slide we create we should consisted of 3 parts in Eash slide " slicer part where should contained a drill through and tooltips, and the cards part, and the main part where contain the charts and graphs" and we also, make sure in this phase we applied all applicable measures and analysis in this part like DAXs functions and

Aggregations functions (sum, count, count rows, distinct count, AV, MIN, MAX), iterations functions (Sumx, Countx, AVx, MINx, MAXx), if, related and switch functions.

The fifth phase is **Testing & Quality Assurance,** and making sure all charts will contain an accurate date to be analyzed, and finally, to make an indicator and benchmark results, we applied the pivot tables in the database Excel date sheet to make it a reference to our results, which we extract from the Power BI.

Project Documentation Draft:

1. PROJECT PLANNING & MANAGEMENT:

1.1 Project Proposal:

This project aims to design and implement an operational HR analytics dashboard using Microsoft Power BI to analyze employee attrition, turnover rate, satisfaction levels, workforce performance, and related organizational factors affecting employee retention.

The project objective is to investigate the root causes of turnover in the HR department, analyze correlations between attrition and contributing factors, and transform raw employee data into interactive business intelligence dashboards to support executive decision-making.

Project scope includes:

- Data collection and validation from the company database and archives.
- Data cleaning and transformation through Power Query.
- Data modeling and KPI development using DAX measures.
- Interactive dashboard development.
- Benchmarking and validation through Pivot Tables and comparative indicators.
- Data reporting.

GANTT CHART (PROJECT PLAN):

Phase	Duration	Owner
Data Collection & Data Transformation	Week 1-2	Team
Data Cleaning & ETL	Week 3	Team
Data Modeling & Relations	Week 4	Team
Data Profiling & Exploration DAX Development	Week 5	Team
Data Visualizations & Dashboard Design	Week 6	Team
Testing	Week 7	Team
Data Reporting	Week 7	Team
Presentation	Week 8	Team

1.2 Team Collaboration and Roles:

Our team consists of six members collaborating to complete the project to the highest standard through distributed responsibilities.

Roles include:

- Data acquisition and liaison with the database/archive department.
- Data cleaning and preparation.
- Power Query transformations.
- Data modeling and DAX development.

- Dashboard visualization and UI/UX structuring.
- Validation, benchmarking, documentation, and presentation.
- Data reporting.

Task allocation ensures resource optimization and accountability throughout all project stages.

1.3 Project Methodology and Milestones:

The project was structured in the following phases:

Phase 1 — Data Collection & Gap Closure:

Contacting the responsible person in the database and archive department to obtain complete datasets as a premium initial phase, ensuring all gaps and missing data were fulfilled before proceeding with analysis.

Deliverables: (Complete HR dataset - Data quality verification - Gap analysis closure).

Phase 2 — Problem Definition & KPI Framework:

Determining that the project requires applying an operational Power BI dashboard methodology, using all applicable KPIs directly related to the core issue:

(Employees KPIs Rate - Attrition Rate - Satisfaction Indicators - Retention Measures - Performance Metrics - Risk Indicators).

Phase 3 — Relationship Analysis:

Analyzing relationships between attrition and influencing factors such as:
(Employee satisfaction - Performance ratings - Department - Age - Management rank - Experience - Compensation - Educational level). A dedicated date table was created, this enables time-based analysis (D, M, Y) trends, the date table was linked to the main dataset.

Phase 4 — Dashboard Development:

Building a four-slide interactive dashboard covering:

1. Attrition Analysis.
2. Employee Satisfaction Analysis.
3. Employee Characteristics & Workforce Analytics.
4. Company Overview & Red Alert Indicators.

Each slide contains:

- Slicer Section (including drill-through and tooltips).
- KPI Card Section.
- Main Visualization Section.

1.4 Risk Assessment and Mitigation:

Potential Risks	Mitigation
Missing or incomplete data	Archive/database validation phase
Poor data quality	Multi-stage Power Query cleansing
Incorrect KPI calculations	Benchmark validation via Pivot Tables
Model relationship issues	Structured data modeling validation
Misinterpretation of trends	Cross-checking with reference indicators

1.5 Project KPIs (Success Metrics)

Project success will be measured using:
(Dashboard accuracy rate - Data completeness ratio - KPI calculation precision - Decision-support usability - Executive reporting effectiveness).

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2. LITERATURE REVIEW:

2.1 HR Analytics Background:

The project adopts principles from:

- Workforce Analytics
- Attrition Modeling
- Employee Satisfaction Analysis
- Business Intelligence Dashboard Methodologies

The literature supports turnover as a multi-factor phenomenon influenced by organizational, behavioral, and compensation-related drivers.

2.2 Suggested Improvements:

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2.3 Feedback and Evaluation:

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3. REQUIREMENTS GATHERING:

3.1 Stakeholder Analysis:

Key stakeholders: (HR Department - Senior Management / CEOs - Database & Archive Department - Employees (indirectly) - Dashboard End Users).
will include:(Accurate attrition monitoring - Predictive warning indicators -Workforce performance visibility - Executive reporting support).

3.2 Functional Requirements:

The system: (Import and transform HR data - Calculate dynamic KPIs - Support drill-through analysis - Generate interactive dashboards - Provide benchmarking indicators - Support alerts and red-flag monitoring).

3.3 Non-Functional Requirements:

(Accuracy -Scalability -Usability -Reliability -Visualization responsiveness).

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4. SYSTEM ANALYSIS & DESIGN:

4.1 Problem Statement & Objectives:

- Main problem:(High turnover rates & lack of integrated analytical visibility into underlying causes).

- Objectives:(Analyze turnover causes - Measure relationships among attrition drivers - Support proactive HR decision-making).

USE CASE DIAGRAM & DESCRIPTIONS (System Actors):

Actors: (HR Analyst - HR Manager - CEO / Executive Management - Database Administrator - System User (Dashboard Viewer).

Use Case Structure for HR Analytics System:

HR Analyst --> (Import Employee Data - Clean & Transform Data - Build KPIs Measures - Generate Dashboards).
HR Manager --> (Monitor Attrition - Review Satisfaction Metrics - Analyze Risk Indicators).
CEO ---> (View Executive Dashboard - Review Alerts & Trends).
Database Admin ---> (Maintain Data Sources - Validate Data Integrity).

Use Case Descriptions:

Use Case 1: Monitor Attrition

Actor: (HR Manager).
Description:(Allows HR management to track turnover trends, analyze attrition causes, and identify risk areas).
Inputs:(Leave dates - Departments - Satisfaction scores).
Output:(Attrition insights - Risk indicators).

Use Case 2: Executive Monitoring

Actor: CEO.
Purpose: (View summarized KPIs and red-alert indicators for decision making).
Outputs: (Turnover Rate - Retention Indicators -Workforce Overview).

Software Architecture:

consists of: (Excel Database Source - Power Query ETL Layer - Data Modeling Layer - DAX Measures Engine - Interactive Dashboard Layer).

4.2 Database Design & Data Modeling:

Employee dataset includes: (Employee IDs - Demographics -Hiring/Leaving Dates - Departments - Performance Ratings - Compensation - Satisfaction Levels). The data model uses relational logic for analytical measures.

ER Diagram:

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Logical and Physical Schema:

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4.3 Data Flow & System Behavior:

Process flow: (Raw Data (Cleaning, Transformation, Modeling, KPI Measures, Dashboard Visualization, Benchmark Validation, Data Reporting).

4.4 DATA FLOW DIAGRAM (DFD):

Context-Level DFD (Level 0):

(Employee Database (Interactive Dashboard - KPI Reports - Executive Alerts).

Level 1 DFD:

(Raw Data -Data Collection - Data Cleaning -Transformation -Data Modeling -KPI Calculations -Dashboard Visualization - Data Reporting - Management Decisions).

ACTIVITY DIAGRAM:

(Start - Acquire Data - Clean Data - Transform Data - Build Relationships - Create DAX Measures -Generate Dashboard -Validate Results -Present Insights -End).

SEQUENCE DIAGRAM:

(User - Requests Dashboard View - Dashboard - Retrieves Data Model - Calculates KPIs - Displays Visualizations - User Reviews Insights).

CLASS DIAGRAM (Conceptual):

Employee: (Employee ID - Name - Age - Hire Date - Leave Date - Salary).
Department: (Dept Name).
Performance: (Rating - Satisfaction).
Relationships: (Employee belongs to Department - Employee has Performance- Employee may have Leave Reason).

DEPLOYMENT DIAGRAM:

(User Devices - Power BI Service - Power BI Desktop Model - Excel HR Database).

COMPONENT DIAGRAM:

(Excel Data Source - Power Query ETL - Data Model - DAX Engine - Power BI Dashboard).

4.5 ENTITY RELATIONSHIP DIAGRAM (ERD):

Since you mentioned mostly one core table plus analytical modeling, a star schema works best.

Fact Table: (Fact Employees (Employee ID (PK) - Hire Date -Leave Date - Salary -Bonus -Satisfaction - Performance)

Dimension Tables:

Dim Department:(Department Name**).**
Dim Employee:(Employee ID -Gender -Age -Marital Status - Education**).**
Dim Date: Date -Year -Month - Quarter**).**
Dim Leave Reason:(Reason ID - Leave Reason - Leave Category**).**

Star Schema:

```
        Dim Date
           |
Dim Dept – Fact Employees – Dim Leave Reason
           |
        Dim Employees
```

4.6 UI/UX Design & Prototyping

- Dashboard wireframe structured into: (Slicers section -KPI cards section - Main analytical visuals)
- Includes: (Drill-through navigation - Tooltips - Alert indicators -Executive summary cards)

4.7 DASHBOARD WIREFRAME MOCKUPS:

Repeated across all 4 slides: (Slicers - KPI Cards - Main Charts / Graph - Risk Alerts / Executive Insight Box)

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5. IMPLEMENTATION:

5.1 Data Preparation in Power Query:

Applied formal and theoretical data preparation steps:

Cleaning Phase: (Handling errors -Replacing values - Removing blank columns -Removing blank rows -Removing duplicates -Using first row as headers - Removing unnecessary top rows)

Transformation Phase: (Correcting data types -Standardizing date formats - Applying local time formatting - Renaming headers -Merging employee name columns - Fill-down techniques for unmerged fields)

Modeling Phase: (Data modeling and relationship creation between tables). A dedicated date table was created. This enables time-based analysis (daily, monthly, yearly trends, the date table was linked to the main dataset.

DAX and Measures Applied: -

- **Aggregation Functions:** (Sum - Count - Count rows - Distinct count - Average - Min - Max)
- **Iterator Functions:** (SUMX - COUNTX - AVERAGEX -MINX -MAXX)
- **Logical Functions:** (If - Switch - Related)
- **KPIs Measures:** (Turnover Rate - Retention Rate -Attrition Metrics - Satisfaction Metrics - Compensation Indicators - Red Alert Indicators)

5.2 Benchmark Validation:

Pivot tables were applied to the Excel database as reference benchmarks to validate extracted Power BI results. "This served as indicator control and result verification".

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6. TESTING & QUALITY ASSURANCE:

Testing activities included:

- KPI validation testing.
- Data accuracy checks.
- Relationship integrity testing.
- Dashboard functionality testing.
- Cross-validation against benchmark pivot results.

Test Cases: -

To make sure all charts contain an accurate date to be analyzed, and finally to make an indicator and benchmark results, we applied the pivot tables in the database Excel date sheet to make it a reference to our results, which we extract from Power BI.

Test Case 1:

Scenario: (Validate turnover rate formula). Expected: (Matches pivot table benchmark).
Result: (Pass).

Test Case 2:

Scenario: (Check blank leave dates excluded from leavers). Expected: (Active employees not counted).
Result: (Pass).

Bug Reports: -

Employee performance should be applied with numbers and percentages to make the indicators more realistic and accurate, to be measurable practically.

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7. FINAL PRESENTATION & REPORTS:

Dashboard Slides:

Slide 1 (HR Employees KPIs):

We have a 694 employees the total active employees is about 592 with percentage of 95% and they are distributed into five departments and the attrition rate is equal 15% and the female employees count is consider 312 employees and the males is 382, besides the managerial level from them is equal 91 and the non-managerial level is equal 603 almost of them are working on production department and they are around 448 And also we figure the resignations number is equal 62 employees and the terminated employees by HR department is count 36 also, The exceeded performance employee is equal 87 in four departments Almost of them are working in production department the female employees count is consider 34 employees and the males is 53, besides the managerial level from them is equal 18 and the non-managerial level is equal 69.

In conclusion: we must focus on the production employees' needs due to the main power on our company concentrated in this department.

Slide 3 (Satisfaction):

We have 694 employees and the total active employees is about 592 with percentage of 95%. the Highest Satisfaction Employees is equal 60.2% and by Age Group are between 40–49 recorded on the satisfaction rate. With 66%. This indicates that mid-career employees tend to have greater job stability and engagement. The satisfaction rate among female employees reached **64%**. This reflects a relatively an intermediate level of satisfaction within the female workforce. According to distribution by department the Z-Manage Department achieved the highest satisfaction level among all departments with 81.25%, And This perhaps indicate effective management practices or a positive work environment within that department. According to the years of experience the Newly hired employees showed the lowest satisfaction levels, and we consider that due to challenges in onboarding, adaptation, or initial expectations for the first year of work. Otherwise, Employees with 2–3 years of experience recorded the highest satisfaction rate. This suggests that employees at this stage are more settled and engaged in their roles and handling their duties and responsibilities with ease. And the most satisfied employees branch who working in buenos Aires with 64.4%.

In conclusion: we must focus on the IT/IS employees' needs besides the junior employees Goals; we must do more efforts and more concentration especially in the first year to gain their satisfaction.

Slide 3 (Attrition KPIs):

The total active employees is about 592 with percentage of 95% and the in active employees is around 102 with a percentage attrition rate 15% concentrated in production department with a 80 employees only in this department leave the company most of them is leaving according to performance, salaries reason and they are about 31 person from them besides the non-managerial level is represent 90.2% of the attired employees and almost of them are female employees with a count of 47 female. Besides that, we figure out from the analysis that the majority of employees leave in the first year of work approximately 60 employees according to performance reason in addition to there are 31 employees attended to leave according to the satisfaction rate less than 3 in the rate ranking. Which they cost the company more than 8.1M $ in the past 3 years from 2017-2019

In conclusion: we must focus on the non-managerial production employees' needs due to the main power on our company concentrated in this department especially for the female section and enhance their salaries also, strength their training once they join the company in the first year to boost their performance evaluations, and looking for the 31 employees who represent high risk people to leave the company very soon .

Slide 4 (Overviews):

We have 694 employees and the total active employees is about 592 with percentage of 95%. And the Attrition rate is still under control, but we have to focus on the production department training and enhance our system to monitored their performance properly and increase their productivity, in aligned with take care of IT/IS department to made their employees more satisfied especially the new comers employees in the non-managerial level to cut-down the cost attrition and shrink the rate of turn over .

Challenges and Outcomes:

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User Manual:

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RED-ALERT INDICATORS (Future Risk Panel):

High Attrition Alert = IF ([Turnover Rate] > .15,"Warning","Normal")

Low Satisfaction Alert = IF ([Avg Satisfaction] < 3,"Critical","Stable")

Technology Stack:

(Microsoft Excel - Microsoft Power BI - DAX -Power Query - GitHub Repository)

Final Deliverables:

(Source files - Power BI Dashboard - Documentation - Presentation Slides - Benchmark Validation Sheets)

GITHUB BRANCH STRATEGY:

(Main, develop, feature-data-cleaning, feature-Dax, feature-dashboard, documentation).

Recommended GitHub Repository Structure:

(Documentation, Dataset, Power BI, Images-Wireframes, Testing, Presentation, README.md).